Practical problems with equations

1 2

2

a $x + y = 13$

b 8 cm

3 3500 people

4 $x = 18$

5 18

6 48 hours

7 $\frac{36}{5}$ hours

8

a 2

b 9

9 10 minutes

10 4

11

a 8 mph

b 16 mph

12

a

	Distance (miles)	Rate	Time
Upstream	15	$n - 3$	$\frac{15}{n-3}$
Downstream	30	$n + 3$	$\frac{30}{n+3}$

b $n = 9$

13 $n = 81$

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14

a 45 mph

b 66 mph

15 23 mph

16

a $(1-k)\,t$

b

i $d = 380kt$

ii $d = 350\,(1-k)\,t$

c $k = \dfrac{35}{73}$

d 1184 miles

17

a $x = 7$

b $x = 3$

c $x = 2$

18

a $m + 71 = 403$

b $m = 332$

19

a $p = 164$

b \$436

20

a $x + 2$

b $x + 4$

c $2x + 4$

21 $10, 12, 14$

22 $13, 15, 17, 19$

23

a $\dfrac{1}{4}x = 3x - 22$

b $x = 8$

24

a $189 = 21x + 105$

b $x = 4$

25

a $24x + 3$

b $x = 13$

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26

a $2x - 3$ **b** 9 inches **c** 15 inches

27 $14, 15, 16$

28

a $x = \dfrac{74}{9}$ **b** $\dfrac{296}{9}$

29 $x = 6$

30

a $x = -1$ **b** 8°

31 $x = 12$

32 $x = 5$

33

a $x = y + 1.36$ **b** $8x + 9y = 119$ **c** $y = 6.36$ **d** $x = 7.72$

34

a $x = y - 4$ **b** $2(x - 4) + 2y = 12$

c $y = 5$ **d** $x = 1$

35

a $x = 7y$ **b** $x + 2 = 5\,(y + 2)$

c $y = 4$ **d** $x = 28$

36 $11\,^\circ\text{C}$

37

a 4 **b** 60

38

a \$271.12 **b** Underestimate **c** \$1.73 **d** 32 years

e 2030

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39 $B = \$14\,736.84$

40 17 years old

41 $36\,000$ kg

42 28 advertisements

43 7 nights

44 240 L

45 10 years old

46

a 8 **b** 81

47 4 hours

48 $2w^2 + 13w = 260$

Additional questions

49

a $x = 2$ **b** $m = -3$ **c** $p = 4$ **d** $k = -9$

50

a $m = 7$ **b** $k = 5$ **c** $t = 1$ **d** $d = -7$

e $f = 4$ **f** $r = -1$ **g** $w = -3$ **h** $p = -5$

51

a $x = 8$ **b** $x = 20$ **c** $x = 75$ **d** No solution

52

a $h = 4$ **b** $x = 4$ **c** $x = 7$ **d** $m = 2$

e $x = -3$ **f** $p = -2$ **g** $k = 12$ **h** $x = 20$

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i $x = 16$ j $x = 11$ k $x = 4$ l $y = -3$

m $x = 70$

53

a $x = -5$ b $x = \frac{2}{5}$ c $x = -\frac{22}{19}$ d $x = \frac{11}{4}$

e $k = 8$ f $y = 8$ g $x = 22$

54

a One solution b Infinitely many solutions

c Infinitely many solutions

55 $x = 3$

56 $x = 4$

57 $x = 16$

58 $x = 13$ inches

59 $p = \$186$

60 $k = 14$

61

a Let x be the exact number of people at the concert. Stormie's guess was 800 too many, so Stormie's guess was $x + 800$. Pierce guessed 300 too few, so Pierce's guess was $x - 300$. The average of their guesses is 4250, so that means $\frac{\text{Stormie's guess} + \text{Pierce's guess}}{2} = 4250$. We can then substitute in the equations we have found for Stormie's and Pierce's guesses to get the equation $\frac{x + 800 + x - 300}{2} = 4250$.

b 4000

62

a $12 = 0.5 \cdot L$
Where L is the number of laps.

b $L = 24$

c Assuming both of them used their own cart, they would need to pay $\$2$ as a deposit. So the equation would instead be $12 = 0.5 \cdot L + 2$

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63 Bassam should have multiplied by $\frac{1\text{ mi}}{1.6\text{ km}}$ to convert from km to miles.

64

a True. If you substitute the solution into the unknown variable and calculate each side separately, you should end up with the same value on both sides.

b False. For example, the equation $x + 1 = 3$ has the solution $x = 2$, and the equation $x - 1 = 1$ also has the solution $x = 2$.

c False. If the variable terms do not cancel out, then the equation must mathematically have one solution.

d True. For example, if a value of x satisfies $2x + 3 = 1$, then it will also satisfy $4x + 6 = 2$.

65

a

i $3n + 3 = 4n + 6$ **ii** One solution

b

i $2n + 6 = 2n + 5$ **ii** No solution

c

i $4n + 2 = 4n + 2$ **ii** Infinitely many solutions

66 $y = 8$

67 $B = \$15\,652.17$

68 $y = 16$ years old

69 $x = 12$

70

a $x = 17$

b $17, 18, 19$

c No, because the equation used to solve for x in part (a) has only one unknown, x, to the power of one. Therefore there is only one solution.

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